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Compact and high-performance polarization beam splitter based on triple-waveguide coupler

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ABSTRACT

Polarization beam splitter (PBS) is a fundamental component of integrated photonics to manipulate wave polarization state, and most PBS devices were designed on the popular SOI. As an alternative, a PBS may be implemented on the basis of thin-film $LiNbO_3$-on-insulator (LNOI), a promising platform for developing various compact or ultra-compact photonic active and passive devices. However, smaller modal birefringence of the LNOI waveguide makes it not easy to implement a PBS of high-performance. Here, we propose a compact, high-performance PBS on the basis of a triple-waveguide coupler consisted of two LNOI ridge waveguides and an amorphous silicon (α-Si) nanostrip assisted waveguide. By making use of the α-Si nanostrip, the PBS works in the case that the TM polarization mode meets the phase matching condition while the TE polarization mode does not. Numerical results show that the PBS is compact with a length of only 31 μm. The device can achieve an ultra-high polarization extinction ratio (>29.4 dB) and an ultra-low insertion loss (<0.066 dB) in the operating wavelength range of 1500–1600 nm for the TE polarization mode, and a polarization extinction ratio of > 20 dB and an insertion loss of < 0.16 dB over a ~ 60 nm bandwidth from 1527 nm to 1587 nm for the TM polarization mode. Moreover, the device has a larger feature size that facilitates fabrication. Overall, this device has the advantages of compact size, higher polarization extinction ratio, lower insertion loss, and larger feature size, which is suitable for on-chip applications.

1. Introduction

Recently, thin-film lithium-niobate-on-insulator (LNOI), which consists of a sub-micrometer-thick lithium niobate (LN) device layer, buried silica (SiO_2) layer, and LN or Si substrate, has already become a photonic platform and received much attention in both classical and quantum photonic integrated circuits (PICs) owing to its good properties including excellent electro-optic and nonlinear optical properties, wide transparency window (0.35–5 μm), and tight optical confinement [1,2]. By utilizing these properties, various compact or ultra-compact photonic devices with excellent performance have already been developed. These include electro-optic modulators [3,4], frequency converters [5,6], lasers [7,8], detectors [9], polarizers [10] and optical switches [11], making the LNOI platform very potential for the PICs.

As we know, the LN exhibits intrinsic birefringence, which is absent in Si-on-Insulator. This intrinsic property results in that an LNOI-based device is polarization-dependent. In addition, electro-optic and nonlinear coefficients of LN are also polarization-dependent, resulting in that the TE- and TM-polarized waves propagating along the LNOI waveguide have different electro-optic and nonlinear effects. Therefore, it is necessary to implement polarization manipulation and management in the LNOI photonic integrated platform. Polarization beam splitter (PBS), which enables separation or combination of two orthogonal modes, is one of the basic building blocks for polarization manipulation and management. It has been extensively applied in coherent optical communication, polarization division multiplexing, and quantum information processing systems [12–14]. Generally, a waveguide with ultra-high mode birefringence facilitates to realize a PBS with outstanding performance [15]. For example, a standard silicon-on-insulator single-mode waveguide with a cross-section size of 220 nm (depth) × 500 nm (width) exhibits modal birefringence >0.7 at 1550 nm wavelength [15]. The quite large modal birefringence makes it easy to implement a PBS. Unfortunately, the modal birefringence of the LNOI single-mode waveguide is relatively low, ~0.2 for an LNOI with a thin-

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film thickness of 500 nm, making it relatively difficult to realize PBS in the LNOI platform. Up to now, just a few structures, including Mach-Zehnder interferometer, stimulated Raman adiabatic passage, metamaterial, and directional coupler, have been proposed and demonstrated to achieve PBS in the LNOI photonic integrated platform [16–20]. The Mach-Zehnder interferometer and the stimulated Raman adiabatic passage structures are the most promising strategy to achieve PBS with broadband (> 100 nm) and high polarization extinction ratio (PER > 20 dB) in the LNOI platform [16,17]. However, these PBSs always have a comparatively larger footprint (typically thousands of square micrometers). Deng *et al*. have designed a broadband (> 85 nm) PBS with a PER of > 20 dB in the LNOI platform using hetero-anisotropic metamaterial by engineering the dispersion and birefringence [18]. Unfortunately, this type of PBS has a complex structure that requires ultra-high-precision fabrication technology. The two-waveguide directional coupler is a relatively simple and promising structure to realize PBS [19,20]. In Ref. [19], a PBS with acceptable performance has been demonstrated based on the symmetric directional coupler. Nonetheless, the total length of the device is up to 85 μm, which makes it challenging to achieve high-density integration. Utilizing asymmetric directional couplers formed by amorphous silicon (α-Si) assisted waveguide, Zhang *et al.* have achieved a compact PBS with low insertion loss (IL < 0.5 dB) [20]. Nevertheless, the α-Si-assisted waveguide is not compatible with the typically LNOI waveguide. Other schemes, such as subwavelength structures and inverse design, are ways to realize ultra-compact and high-performance PBS and have been successfully demonstrated in the silicon-on-insulator photonic platform [21–23]. These schemes, however, may not be suitable for the LNOI platform due to the difficulty of fabricating structures with feature sizes of tens of microns caused by the inherent chemical stability of the LN. Consequently, it remains challenging to implement a compact and high-performance PBS in the LNOI photonic integrated platform.

The triple-waveguide coupler is an alternative solution for realizing PBS. In general, this structure of PBS can achieve a higher PER with comparable operating bandwidth, IL, and size to the two-waveguide coupler-based PBS [24,25]. In this paper, we propose a compact and high-performance LNOI-based PBS based on an α-Si nanostrip-assisted triple-waveguide coupler. The coupler contains an α-Si-assisted waveguide in the centre and two LNOI ridge waveguides on the two sides. Although the involvement of the α-Si nanostrip makes the fabrication of the device more complicated, it provides a new degree of freedom to manipulate the polarization state, leading to the ability to overcome the drawbacks of low mode birefringence in the LNOI waveguide. With the aid of the α-Si nanostrip, the TM polarization mode can achieve phase matching, while the TE polarization mode exhibits a significant phase mismatch. The results show that for the TE polarization mode, the PBS achieves an ultra-high PER (> 29.4 dB) and an ultra-low IL (< 0.066 dB) in the operating wavelength range of 1500–1600 nm. For the TM polarization mode, the PBS achieves a PER of > 20 dB and an IL of < 0.16 dB in the operating wavelength range of 1527–1587 nm. In particular, the PER of PBS is up to 55 dB at 1551.6 nm wavelength. Moreover, the device has a compact size of 31 μm in length and is well suited for use in PICs. To the best of our knowledge, such a compact size, high-performance PBS has not been achieved yet. Therefore, the proposed PBS is expected to offer a promising novel route for achieving polarization manipulation in LNOI-based PICs.

2. Device structure and principle

Fig. 1(a) and (b) show the three-dimensional (3D) schematic diagram of the proposed PBS and the cross-section view in the *xz* plane of the triple-waveguide coupler, respectively. The PBS consists of an input port, a through port, a cross port, an S-shaped bent waveguide, and a triple-waveguide coupler. The triple-waveguide coupler with length L_c consists of two LNOI ridge waveguides (WGA and WGB) with width W_1 and a hybrid waveguide (WGC) formed by loading α-Si nanostrip on the top center of a narrow LNOI waveguide with width W_2. The width and thickness of the α-Si nanostrip are defined as W_{Si} and H_{Si}, respectively. The α-Si can be fabricated on LNOI waveguides using plasma-enhanced chemical vapor deposition method [26,27]. The waveguide gap between WGA and WGC, and WGB and WGC, is fixed at $G = 200$ nm, which is easy for fabrication. The S-shaped bent waveguide is designed at the end of WGA to decouple the through and cross ports. The offset of the S-shaped bent in the x and y directions are $L_x = 1.5$ μm and $L_y = 8$ μm, respectively. The input, through, and cross ports have the same width (W_1) and are consistent with WGA and WGB. In our simulations, the sidewall angle (θ) of the LNOI ridge waveguide is fixed at $\theta = 83°$ to match the state-of-the-art focused ion beam process technology [28,29]. The proposed PBS is designed based on a Z-cut LNOI wafer with a 500-nm-thick (H_{LN}) LN device layer and a 4.7-μm-thick (H) SiO_2 buried layer. In principle, either an X-cut or a Z-cut LNOI can be used to design a PBS based on the proposed structure (in particular, for the X-cut/Z-propagation scheme, calculations show that the TM and TE modes have still a significant anisotropy of ~ 0.12, which is comparable to that of Z-cut/X (Y)-propagation scheme, ~0.16. So, a PBS can be also designed using the X-cut/Z-propagation scheme). Here, we exemplify a Z-cut LNOI wafer to carry out the design of the proposed device. The thickness of SiO_2 layer is chosen on the basis of its effect on the evanescent field distribution in the layer and hence the waveguide loss. Analysis of waveguide properties shows that the 1/e penetration depth of evanescent wave in the SiO_2 layer is similar to 1.5 μm. This means that the loss figure caused by the evanescent field can be neglected as long as the SiO_2 layer is thicker than 1.5 μm. The typical SiO_2 layer thickness ~ 2. Here, a thicker layer of 4.7 μm, which was adopted occasionally, is considered to ensure that the resultant waveguide loss is sufficiently small and ignorable. The medium on the top of the device is assumed as air ($n_{air} = 1$). In this work, the

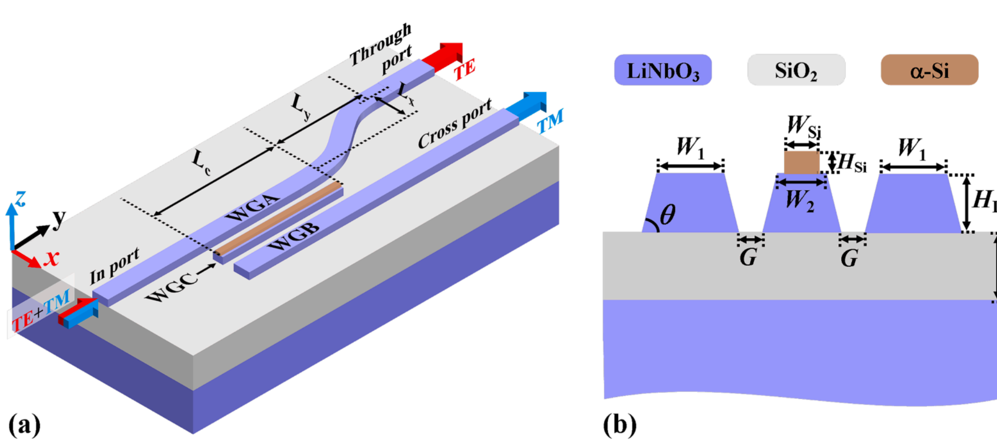

Fig. 1. Schematic diagram of proposed LNOI-based PBS. (a) 3D view and (b) cross-section view in the *xz* plane of the triple-waveguide coupler.

refractive indices of the LN, α-Si, and SiO_2 are taken from the literature [30,31], and [32], respectively. In particular, for the LN, its refractive indices are $n_o = 2.211$ and $n_e = 2.138$ at the 1550 nm wavelength. The working wavelength utilized in the following simulations is set at 1550 nm unless otherwise indicated.

The α-Si nanostrip that loads on the top center of the narrow LNOI waveguide provides a novel dimension to manipulate the mode refractive index. With proper design of the geometry of the α-Si nanostrip, only TM polarization mode satisfies the triple-waveguide phase-matching condition, while the TE polarization mode exhibits a significant phase mismatch. For the TM-polarized wave, it can resonantly couple between the three waveguides, and the coupling length L_{TM} is also relatively short due to the strongly evanescent wave coupling effect [25]. When the length L_c of the triple-waveguide coupler is the same as the coupling length L_{TM}, the TM-polarized wave can be completely coupled from the input port to the cross port. The TE-polarized wave, which does not satisfy the triple waveguide phase matching condition, couples only between the WGA and the WGB. However, the gap between WGA and WGB is $> 2G + W_2$, leading to an extra-long coupling length L_{TE}. Therefore, at a shorter coupling length L_{TM}, the coupling of TE-polarized wave is almost negligible, which means that the TE-polarized wave is almost entirely output at the through port. By taking advantage of the substantial difference in coupling length between TE and TM polarization mode, an effective PBS is achieved.

The coupling mechanism of the PBS can be further explained using the supermodes in the triple-waveguide coupler. Fig. 2 displays the dominant component profile of the electric field of the supermodes supported by the triple-waveguide coupler at 1550 nm wavelength with $W_1 = 0.85$ μm, $W_2 = 0.3$ μm, $W_{Si} = 0.2$ μm, and $H_{Si} = 0.19$ μm. We can see that for the TM polarization, the coupler supports three supermodes, including two symmetric modes and one antisymmetric mode, and the refractive indices of the supermodes are n_{TM1}, n_{TM2}, and n_{TM3}, respectively. When choosing proper waveguide geometry to satisfy the phase-matching condition [25]

$$\Delta n = n_{TM1} + n_{TM2} - 2n_{TM3} = 0, \tag{1}$$

the maximization of wave coupling from the input port to the cross port can be achieved, and the coupling length L_{TM} can be calculated by [25]:

$$L_{TM} = \frac{\lambda_0}{2(n_{TM1} - n_{TM3})}, \tag{2}$$

where λ_0 is the working wavelength. Substituting the refractive index shown in Fig. 2 into Eq. (2), the calculated L_{TM} is ~ 23.4 μm. For the TE polarization, the triple-waveguide coupler supports only one even mode and one odd mode, which is similar to the two-waveguide coupler, and its coupling length can be calculated by [25]:

$$L_{TE} = \frac{\lambda_0}{2(n_{TE1} - n_{TE2})}, \tag{3}$$

where n_{TE1} and n_{TE2} are the indices of even mode and odd mode, respectively. Substituting the refractive index shown in Fig. 2 into Eq. (3), the calculated L_{TE} is about 1550 μm, significantly longer than that of L_{TM}. The above analysis shows that the supermode distribution for TE and TM polarization exhibits a significant difference in the triple-waveguide coupler, which leads to the significant differences between coupling lengths of TE and TM polarization, thus making it easy to achieve PBS.

To show the superiority of the PBS based on a triple-waveguide coupler, we have also calculated the coupling lengths of TE and TM polarization modes guided in a two-waveguide symmetric coupler consisting of WGA and WGB. The gap G between the WGA and the WGB is fixed at 0.2 μm in the calculation. The calculated coupling lengths are 81.94 μm and 28.47 μm for the TE and TM polarization modes, respectively. Further calculations reveal that the coupling length ratio between TE and TM modes is only ~ 3, which is much smaller than that in the triple-waveguide coupler, ~66. In a word, the insertion of the central waveguide, i. e., the use of triple-waveguide coupler structure, enables to get a larger coupling length ratio, which favors to realize a higher PER desired for a PBS [25].

3. Results and discussion

In order to realize the proposed PBS, it is crucial to choose the appropriate waveguide dimension to achieve effective coupling for TM polarization mode and suppress coupling for TE polarization mode simultaneously. Here, we calculate the mode effective refractive indices for the LNOI ridge waveguide at various waveguide widths using Finite-Difference Eigenmode (FDE) solver (Ansys Lumerical MODE Solutions), as shown in Fig. 3. It is clear that both the mode effective refractive index and the number of modes supported by the waveguide increase with the increased waveguide width. We also can see that the cutoff widths for TE and TM guided modes are 0.39 μm and 0.32 μm, respectively. For the proposed PBS, the LNOI ridge waveguide contains two different widths, W_1 and W_2, which are to be determined. Considering the operation principle of the proposed device, we choose the width W_1 to be 0.85 μm, which can not only meet the requirements of the single-mode condition but also provide a strongly evanescent field for the TM mode to enable effective coupling. For another width (W_2), we choose to be 0.3 μm, which can effectively separate WGA and WGB, thus suppressing the direct coupling between these two waveguides. In addition, such a width (W_2) of the LNOI ridge waveguide does not support TE and

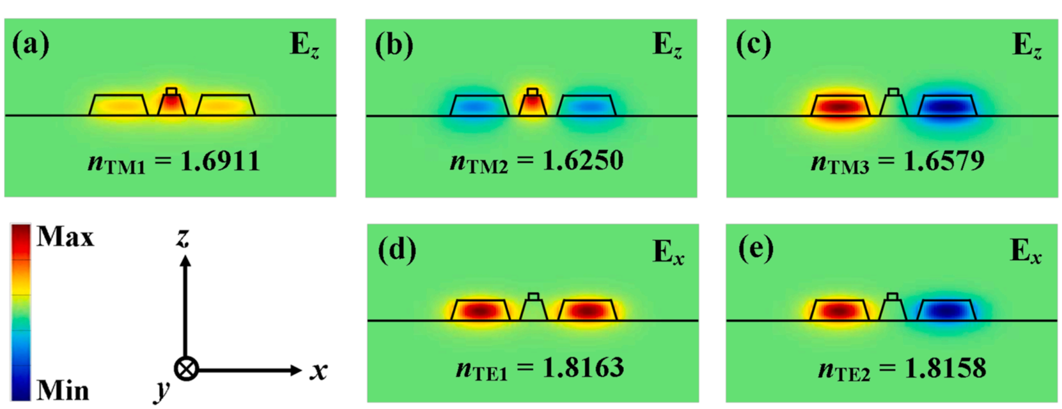

Fig. 2. The electric field distribution of the dominant components for TM (E_z) and TE (E_x) supermodes in a triple-waveguide coupler at 1550 nm wavelength. (a), (b), and (c) represent the 1st symmetric, 2nd symmetric, and antisymmetric supermodes for TM polarization, respectively; (d) and (e) represent the even and odd modes for TE polarization, respectively.

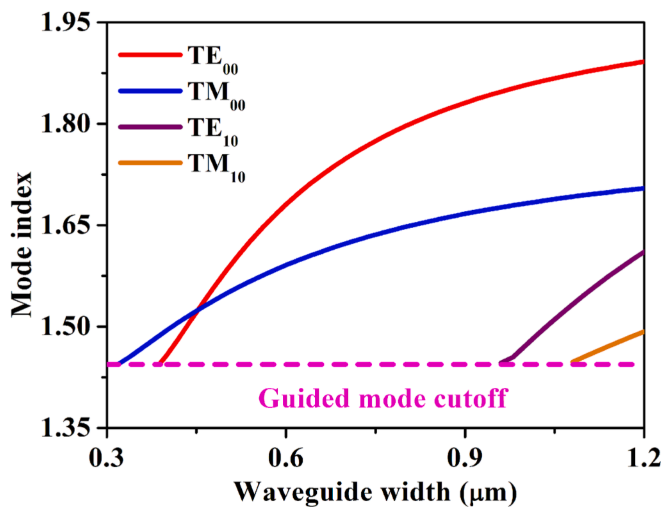

Fig. 3. The effective indices of TE_{00}, TM_{00}, TE_{10}, and TM_{10} modes as a function of the waveguide width of the LNOI ridge waveguide.

TM guided modes, which facilitates the implementation of PBS by employing the α-Si nanostrip-assisted strategy. The reason is that the specific α-Si nanostrip geometry barely affects the TE mode index but can dramatically increase the TM mode effective refractive index [20], which leads to only achieve phase matching for the TM mode.

As mentioned above, it is feasible to satisfy the triple-waveguide phase matching condition for TM polarization mode, i.e., $\Delta n = n_{TM1} + n_{TM2} - 2n_{TM3} = 0$, by modulating the geometry parameters of the α-Si nanostrip. To clarify the effect of the width (W_{Si}) and thickness (H_{Si}) on the phase matching for the TM polarization mode, we calculate the Δn ($n_{TM1} + n_{TM2} - 2n_{TM3}$) with the width varying from $W_{Si} = 120$ nm to $W_{Si} = 300$ nm and the thickness ranging from $H_{Si} = 140$ nm to $H_{Si} = 260$ nm, as illustrated in Fig. 4, where the solid black curve represents the result for $\Delta n = 0$. At the width and thickness of the α-Si nanostrip corresponding to the solid black curve, the TM polarization mode satisfies the phase matching condition and is able to achieve maximum coupling efficiency. From Fig. 4, we can observe that the Δn increases from negative to positive values with the increase of the α-Si nanostrip thickness (width) at a fixed width (thickness). In addition, as the width W_{Si} increases, the thickness H_{Si} corresponding to the maximization of the coupling efficiency decreases. Consequently, on the basis of these findings, we can select the proper geometry parameters of α-Si nanostrip to obtain the highest coupling efficiency for TM polarization mode.

From the above analysis, we have obtained a series of structural

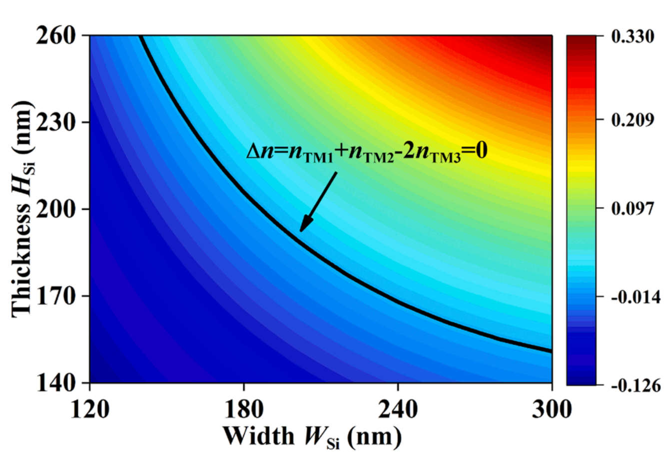

Fig. 4. The results of $\Delta n = n_{TM1} + n_{TM2} - 2n_{TM3}$ with the width varying from $W_{Si} = 120$ nm to $W_{Si} = 300$ nm and the thickness ranging from $H_{Si} = 140$ nm to $H_{Si} = 260$ nm.

parameters of the α-Si nanostrip that enable the TM polarization mode to satisfy the phase matching condition. Nevertheless, not all of them are suitable for the proposed PBS due to not considering the coupling characteristics of the TE polarization mode, and thus further determining the width W_{Si} and thickness H_{Si} of the α-Si nanostrip is required. For this purpose, we calculate the coupling lengths of TM and TE polarization modes at various widths W_{Si} and thicknesses H_{Si} based on Eqs. (2) and (3), respectively, as depicted in Fig. 5(a). Note that the widths and thicknesses used here correspond to the parameters represented by the solid black curve in Fig. 3. For the proposed device, a larger L_{TE} and shorter L_{TM} are preferable because it indicates that TE polarization mode achieves weaker coupling and TM polarization mode achieves complete coupling, resulting in a higher PER. From Fig. 5(a), we can see that the coupling length decreases for both TE and TM polarization modes as W_{Si} increases, which means that the choice of α-Si nanostrip geometry requires a trade-off between L_{TE} and L_{TM}. To analyze this trade-off, we extract L_{TM} and L_{TE} from Fig. 5(a) and calculate L_{TE}/L_{TM} as a function of width W_{Si}, as shown in Fig. 5(b). One can see that the narrower the width W_{Si} is, the larger the L_{TE}/L_{TM} is. It seems to indicate that the width set to $W_{Si} = 120$ nm is the optimal choice, but the corresponding α-Si nanostrip thickness is $H_{Si} = 260$ nm. Such a cross-section of this dimension is challenging for fabrication. In addition, a narrower width W_{Si} is undesirable for compact devices because it leads to a longer coupling length. Hence, we eventually select the width and thickness of the α-Si nanostrip to be $W_{Si} = 200$ nm and $H_{Si} = 190$ nm, respectively, which can provide a relatively large L_{TE}/L_{TM} to ensure a higher PER, as well as a relatively short L_{TM} to assure a compact size. More notably, the final selection of the geometry for the α-Si nanostrip can also reduce the difficulty of the fabrication of the device.

In general, there is a coupling between the S-shaped bent waveguide and cross port, which causes a deviation between the coupling length L_c of the device and the coupling length L_{TM} calculated by Eq. (2). To explore the impact of the S-shaped bent waveguide on the coupling length L_c of the proposed PBS and determine the coupling length L_c, we calculate the coupling efficiency for TM polarization mode with the coupling length varying from $L_c = 21$ μm to $L_c = 26$ μm using the 3D finite-difference time-domain (FDTD) method (Ansys Lumerical FDTD Solutions), as shown in Fig. 6. It can be seen that the coupling efficiency increases initially with increasing coupling length L_c and then decreases. In particular, the TM polarization mode achieves a higher coupling efficiency at the coupling length of about 23 μm. Such a coupling length L_c has a slight decrease compared to the L_{TM} calculated by Eq. (2), indicating that there is very weak coupling between the S-shaped bent waveguide and the cross-port, which further verifies that the choice of 0.3 μm for W_2 can effectively suppress the direct energy exchange between the WGA and WGB. Here, we eventually select a coupling length of $L_c = 23$ μm to achieve a higher coupling efficiency for the TM polarization mode.

The optimized structure parameters of the proposed PBS are summarized below: $L_x = 1.5$ μm, $L_y = 8$ μm, $L_c = 23$ μm, $W_1 = 0.85$ μm, $W_2 = 0.3$ μm, $W_{Si} = 0.2$ μm, $H_{Si} = 0.19$ μm, and $G = 0.2$ μm. Employing these optimal structural parameters, we further analyze the wavelength dependence of the proposed device. For a PBS, the key figures of merit are the PER and the IL, and they are defined as follows [20]:

$$PER_{TE} = 10\log_{10}\left(T_{through,TE}/T_{cross,TE}\right), \tag{4}$$

$$PER_{TM} = 10\log_{10}\left(T_{cross,TM}/T_{through,TM}\right), \tag{5}$$

$$IL_{TM} = -10\log_{10}\left(T_{cross,TM}\right), \tag{6}$$

$$IL_{TE} = -10\log_{10}\left(T_{through,TE}\right), \tag{7}$$

where $T_{through,TE(TM)}$ and $T_{cross,TE(TM)}$ denote the transmittances at through and cross ports when TE (TM) polarization mode is injected, respectively.

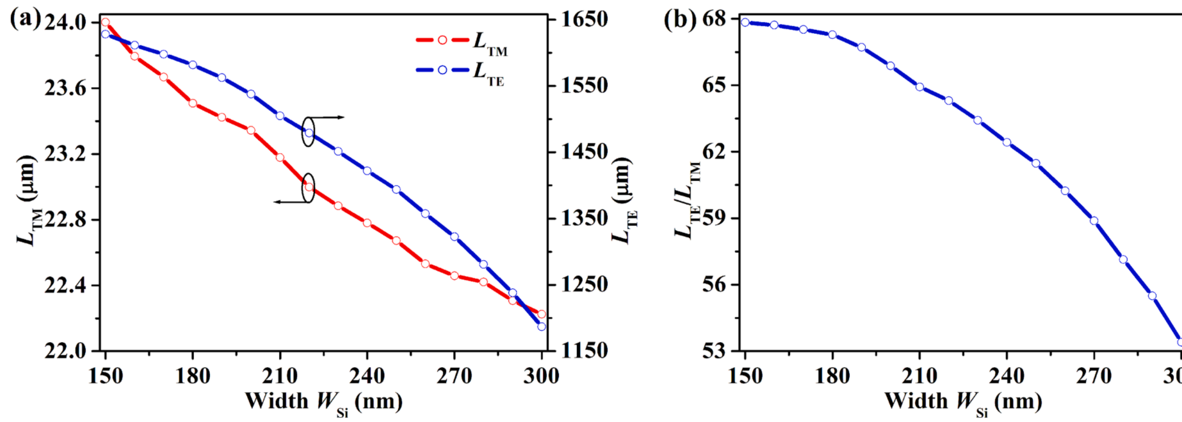

Fig. 5. (a) The coupling length of the TM (L_{TM}) and the TE (L_{TE}) polarizations. (b) The calculated L_{TE}/L_{TM} as a function of width W_{Si}.

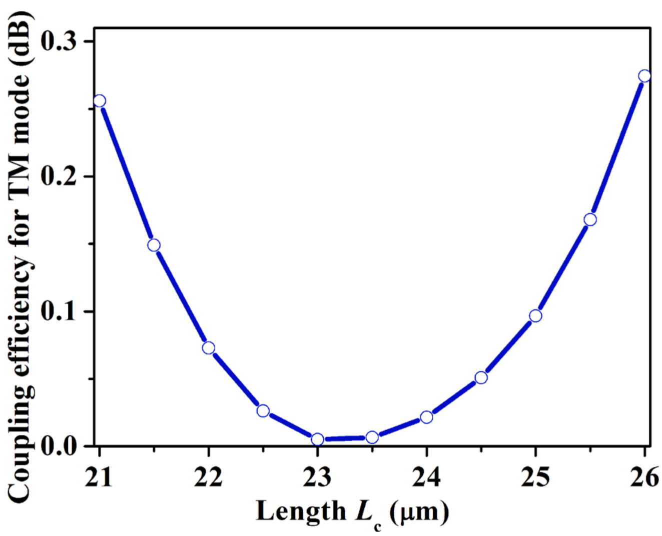

Fig. 6. The coupling efficiency for TM polarization mode as a function of the coupling length L_c.

Fig. 7 shows the PER and IL of the proposed PBS calculated in the wavelength range from 1500 nm to 1600 nm. From Fig. 7(a), we can see that for the TE polarization mode, the PER and IL of the device decrease from 34.8 dB to 29.4 dB and increase from 0.057 dB to 0.066 dB, respectively, as the wavelength increases from 1500 nm to 1600 nm, which indicates that the PBS can achieve a ultra-high PER and ultra-low IL for the TE polarization mode. As to the TM polarization mode, we can observe from Fig. 7(b) that the PER of the PBS is > 11.8 dB for the calculated wavelength range. One can further see that the PBS exhibits an operating bandwidth of about 60 nm (1527–1587 nm) for incident TM polarization mode at a PER of > 20 dB and an IL of < 0.16 dB. Specifically, the device has a PER up to 55 dB at 1551.6 nm wavelength. Moreover, Fig. 7(b) illustrates that the PER of the device for TM polarization mode exhibits definite wavelength sensitivity. To justify the wavelength sensitivity of PER for TM polarization mode, we have calculated the variation of supermode effective indices as a function of wavelength. The results are shown in Fig. 8. One can see that phase matching cannot be realized for the TE polarization mode. For the TM polarization mode, the phase matching can be realized just only at a specific wavelength ~ 1550 nm, i. e., the intersection of the magenta and blue curves, while cannot in a certain wavelength range due to dispersion effect, which causes the wavelength selectivity of PER. Fig. 9 plots the transmission profile of the electric field |E| at the *xy* plane for the TE and TM polarization modes. It can be seen that the incident TE polarization mode propagates directly to the through-port. In contrast, the incident TM polarization mode achieves a strong coupling in the triple-waveguide coupler, leading to the output from the cross port. Such an electric field transmission profile indicates that the proposed PBS effectively separates TE and TM polarization modes with a coupling length of only 23 μm. This implies that the length of our proposed PBS is only 31 μm in total, which includes 8-μm length of the S-bend section. Compared with the reported LNOI-based PBSs [16–19], our proposed device exhibits some advantages, e.g., compact footprint, higher PER, and lower IL.

In practice, the device parameters always deviate from the optimal ones due to defects in the manufacturing process. Therefore, we evaluate the fabrication error tolerance of the proposed PBS, as shown in Fig. 10.

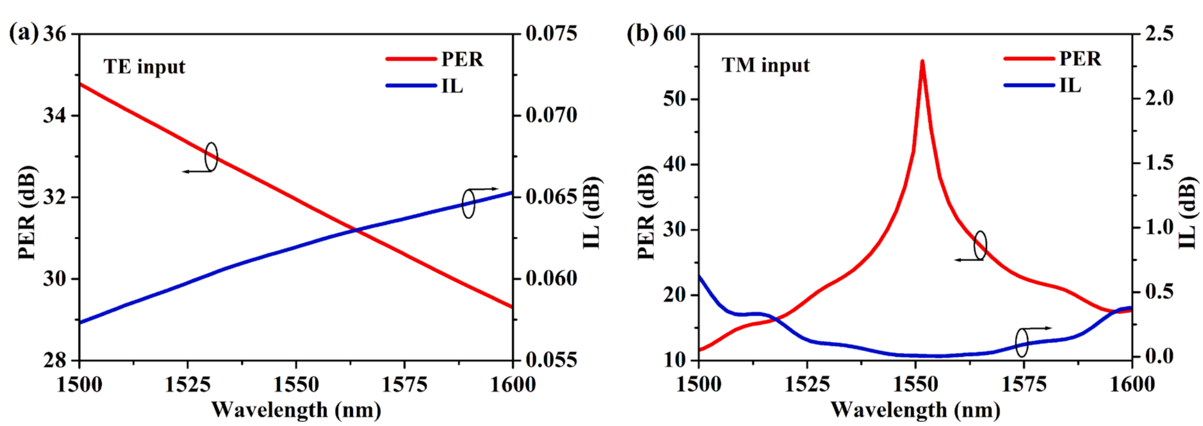

Fig. 7. The PER and IL spectra at the wavelength range of 1500 nm to 1600 nm with incident (a) TE and (b) TM polarized waves.

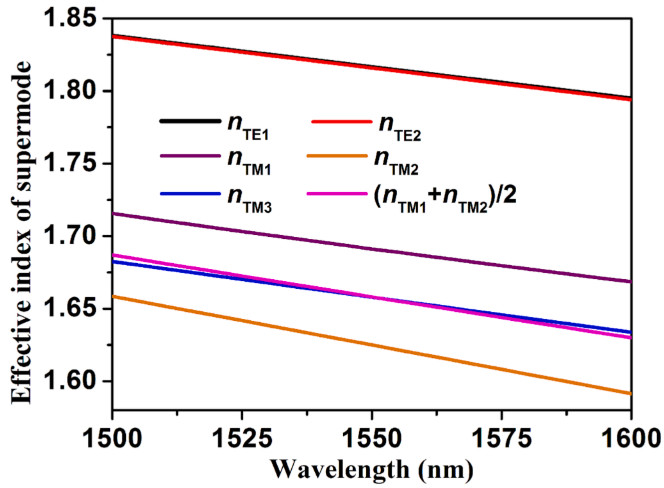

Fig. 8. The variation of supermode effective indices as a function of wavelength in triple-waveguide coupler with the structural parameters of $W_1 = 0.85$ μm, $W_2 = 0.3$ μm, $W_{Si} = 0.2$ μm, $H_{Si} = 0.19$ μm.

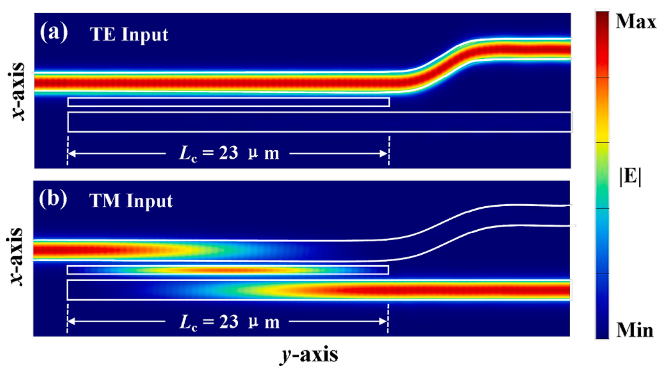

Fig. 9. The simulated propagation profile of the electric field |E| at the wavelength of 1550 nm with the TE and TM polarization modes.

For the proposed PBS, the main fabrication errors include the α-Si nanostrip width (W_{Si}) deviation δW_{Si}, the α-Si nanostrip (H_{Si}) thickness deviation δH_{Si}, the α-Si nanostrip central position deviation δY, and the waveguide gap (G) deviation δG. Here, we assume that both W_1 and W_2 deviation δW when waveguide gap deviation δG, and δG = -δW. From Fig. 10, we can find that for the TE polarization mode, the effect of fabrication errors on the device performance is almost negligible. For the TM polarization mode, the fabrication error only slightly affects the IL of the device. Nevertheless, the PER is relatively sensitive to fabrication errors. In detail, the α-Si nanostrip width deviation, thickness deviation, and waveguide gap deviation causes blue or red shift of the maximum of PER, and they also lead to the reduction of the PER from > 20 dB to > 10 dB in the wavelength range of 1527–1587 nm; the α-Si nanostrip central position deviation reduces the maximum of the PER from 50 dB to 35 dB but does not affect the operating bandwidth with PER > 20 dB. Overall, with the width deviation of $\delta W_{Si} = \pm 5$ nm, the thickness deviation of $\delta H_{Si} = \pm 5$ nm, the central position deviation of $\delta Y = \pm 10$ nm, and the waveguide gap deviation of $\delta G = \pm 5$ nm, the performance of our proposed PBS is still within a reasonable range despite a slight weakening. It indicates that the proposed device exhibits a certain degree of robustness to the fabrication.

4. Conclusion

We have demonstrated a compact and high-performance LNOI-based PBS based on a triple-waveguide coupler formed by two LNOI ridge waveguides and an α-Si nanostrip-assisted waveguide. The intervention of α-Si nanostrip provides a new degree of freedom to manipulate the mode refractive index, enabling solving the problem of a lower mode birefringence suffered by an LNOI waveguide. By tailoring the width and thickness of α-Si nanostrip, the TM polarization mode can meet the phase matching condition and is coupled effectively in the triple-waveguide coupler. Instead, the TE polarization mode does not satisfy the phase mismatch condition, and the coupling is adequately suppressed. The simulation results show that the proposed PBS has a length of only 31 μm, which is more compact than most previously reported LNOI-based splitters and suitable for application in high-density integration. Such a PBS exhibits an ultra-high PER (>29.4 dB) and an ultra-low IL (<0.066 dB) in the operating wavelength range of 1500–1600 nm for TE polarization mode. For the TM polarization mode, although the PBS exhibits definite wavelength sensitivity, it still achieves a PER of > 20 dB and an IL of < 0.16 dB over a ~ 60 nm bandwidth from 1527 nm to 1587 nm. In particular, the PBS has a PER up to 55 dB at 1551.6 nm wavelength for the TM polarization mode. In addition, the presented device has a simple structure and larger feature size (~200 nm), making the fabrication of the device easy. Present work provides a novel solution to implement a compact and high-performance PBS based on the LNOI photonic platform. Such a PBS with outstanding performance has endless potential for application in LNOI-based on-chip coherent optical

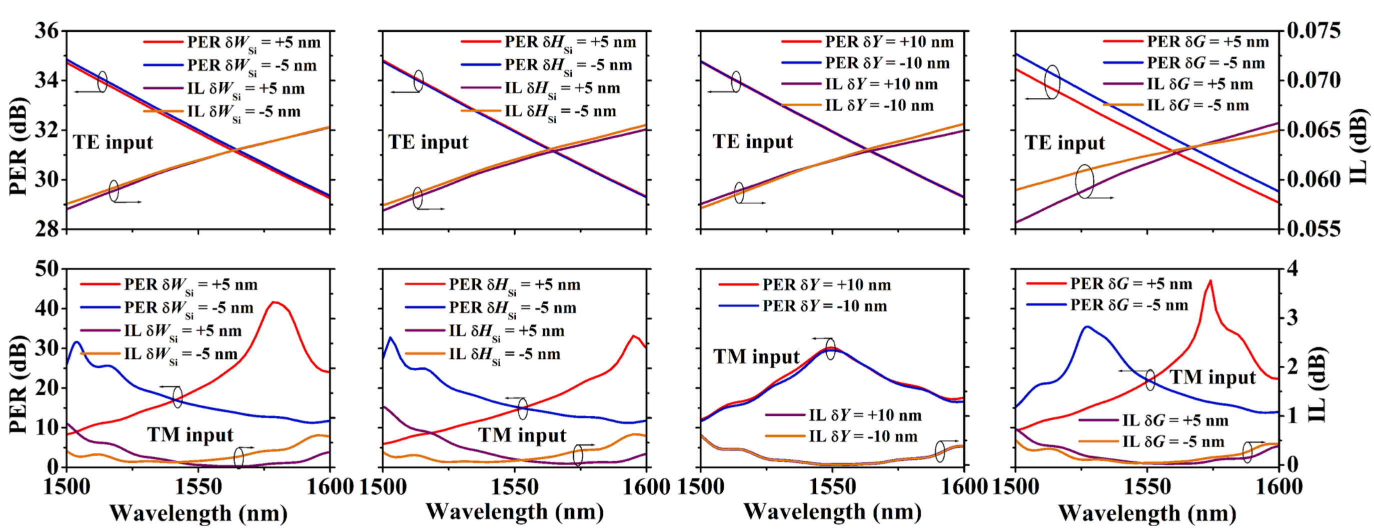

Fig. 10. The calculated PER and IL for the proposed PBS with deviated parameters.

communication, polarization division multiplexing, and quantum information processing systems.

CRediT authorship contribution statement

Jia-Min Liu: Investigation, Software, Writing – original draft. **De-Long Zhang:** Conceptualization, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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